An Explainable, Verifiable Platform for Disaster Aid Request, Tracking and Distribution in Sri Lanka: Design and Multi-Method Evaluation

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***Abstract*—Disaster relief in Sri Lanka is still coordinated largely through phone calls and social media, which makes it hard to see which needs are outstanding, how requests are prioritised, and whether donated goods arrive. Existing platforms address parts of this problem, but few combine open request intake, an inspectable prioritisation rule and verified delivery. Following design science research, we built a web and mobile platform for victims, donors, volunteers and administrators that pairs a disclosed rule-based priority score with priority-first donation matching, a single-use QR handover check and a public, aggregated dashboard. We evaluated the artefact with automated and security-rule tests, concurrent-load tests, a synthetic allocation simulation, and a 19-person usability study. In simulation, priority-first matching served almost all critical needs (99.6% vs. about 79% for first-come or nearest-first rules) but left most low-severity needs unmet (37% vs. 78%), and its distance tie-break decided 72% of matches. Testing led to two fixes. Matching now runs in a database transaction, which removes double allocation but, when 30 matches compete, stretches the matching p95 to between 25 and 271 s. A cache-stampede defect was also removed (p95 about 11 s to 2.4 s). On a temporal hold-out, an auxiliary flood-risk model beat climatology baselines (AUC 0.761 vs. 0.720). Participants gave a mean System Usability Scale score of 76.7 (95% CI 71.0–82.4) and completed 76% of tasks unaided. The platform supports more transparent coordination, but its effect on equitable, timely distribution in a real disaster remains unmeasured.**

***Index Terms*—disaster management, humanitarian logistics, aid prioritisation, explainable decision rules, delivery verification, design science research, Sri Lanka**

I. Introduction

Sri Lanka faces recurrent floods, landslides and cyclones. The May–June 2024 floods affected more than 235,000 people [1], and Cyclone Ditwah in late 2025 caused 646 deaths, left 173 missing and affected between 1.2 and 2.2 million people across all 25 districts (the situation report gives both figures) [2], [3], with damage estimated at US$4.1 billion, about 4% of GDP [4].

Studies of Sri Lankan disaster response describe coordination through phone calls, social media and manual records in the health sector [5] and limited cross-agency collaboration in flood response [6]. Humanitarian-logistics research shows that, without shared information, relief tends to favour accessible areas [7] and that unsolicited donations create "material convergence" that burdens responders [8]. During Cyclone Ditwah, a volunteer platform received about 200 SOS requests representing more than 10,000 people within hours of launch [9], [10], which indicates strong demand for a structured channel.

We ask: *how can a centralised aid request, tracking and distribution system support transparency, coordination and efficiency in Sri Lankan disaster relief?* The contributions are: (1) a four-role web and mobile artefact with explicit design principles; (2) a disclosed allocation rule, analysed for sensitivity, starvation and distance feasibility through simulation; (3) a delivery-verification protocol with a stated threat model; and (4) a multi-method evaluation whose limits are reported alongside its results.

II. Related Work

A. Coordination Platforms

Sahana was developed in Sri Lanka after the 2004 tsunami. It provides organisation, camp and people registries and a request-management module that matches pledges of assistance to requests [11], but it was not adopted nationally in later Sri Lankan floods [12]. Ushahidi-style crowdsourcing surfaces needs quickly [13], and crisis informatics has since added social-media analytics and machine classification of messages [14], [15], but such tools suffer from duplicate and mislocated reports [16] and stop at situational awareness. FloodSupport.org, built during Cyclone Ditwah, collected SOS reports through the web and SMS in three languages [9], [10], but its verification and prioritisation methods are not published. AidTrack [17] verifies pre-registered households with QR codes within one municipality and does not disclose its AI-assisted prioritisation. The official channels of the national Disaster Management Centre (DMC) focus on warnings and situation reports [18].

B. Allocation, Equity and Explainability

Relief allocation balances efficiency, efficacy and equity [19], [7], and last-mile distribution models make explicit how scarce supplies are rationed between locations [20]. Interest in explainable AI in disaster risk management reflects the difficulty of justifying opaque scores to affected people [21]. Blockchain-based donation tracking makes a transaction traceable once a match has been made but leaves the matching decision open [22].

Table I compares named systems on stated criteria. Sahana already integrates requests and pledges. The gap we address is therefore narrower than a missing integrated platform: an

TABLE I
COMPARISON OF NAMED SYSTEMS

Criterion	Sahana	Ushahidi	FloodSup.	AidTrack	Ours
Open request intake	✓	✓	✓	×	✓
Donation/pledge matching	✓	×	?	P	✓
Published priority rule	×	×	×	×	✓
Volunteer dispatch	?	×	P	×	✓
Verified handover	?	×	?	✓	✓
Public aggregate dashboard	?	✓	?	P	✓
Native mobile app	?	?	?	×	✓

✓ = documented in the cited source; P = partial (a related feature exists but does not meet the criterion in full); ? = not documented; × = documented as absent or out of scope.

inspectable prioritisation rule, *verified* handover and *public* aggregate reporting, combined in a mobile-first tool for the Sri Lankan setting.

III. METHOD

We followed design science research [23] using the six DSRM phases [24]. In Gregor and Hevner's terms [25], the contribution is an *improvement*: a new solution to a known problem. Four design principles guided the artefact: **DP1** every allocation decision can be traced to disclosed inputs; **DP2** a delivery closes only with evidence produced at handover; **DP3** public views show aggregates only, never personal data; **DP4** every write passes server-side business rules. Requirements came from secondary sources and the comparison in Section II, giving 33 functional requirements (FR-01 to FR-33). No domain experts were consulted, which is a threat to validity (Section VI-A).

The evaluation followed the FEDS "technical risk and efficacy" strategy [26]: artificial evaluation first (unit, integration and security-rule tests, load tests, simulation), then a naturalistic usability study. Participation was voluntary and anonymised, with written consent. As a low-risk, minimal-data student evaluation, the study was exempt from formal ethics-committee review and was approved by the project supervisor; no personally identifying data were stored.

IV. DESIGN

A. Architecture

Fig. 1 shows the layers. React (web) and Expo/React Native (mobile) clients call one Node.js/Express API, which enforces role middleware and all business rules. Firestore and Firebase Authentication store data. Clients may *read* live data through Firestore listeners, but every *write* goes through the API (DP4), and Firestore security rules repeat the role and ownership checks. External services provide hazard feeds (DMC, GDACS, USGS, Irrigation Department gauges), routing, and parallel push (FCM) and SMS notifications. The API runs as serverless functions on Vercel.

B. Priority Rule

Each request r is scored at submission:

$$S(r) = w_{\text{sev}}(r) + \min(2n_r, 30) + 5v_r + \min(1.5h_r, 20), \quad (1)$$

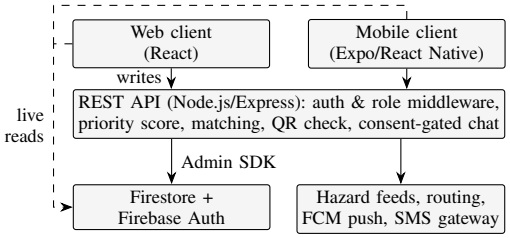


Fig. 1. Architecture. Solid arrows: requests through the API, the only write path. Dashed: read-only Firestore listeners, constrained by security rules.

Algorithm 1 Priority-First Matching (Deployed)

Require: donation d: category c, location ℓ_d, remainder q_d

$C \leftarrow$ pending items of category c in verified requests
sort C by $S(r)$ desc., then $\text{Haversine}(\ell_d, \ell_r)$ asc.
if $C = \emptyset$ **then**
  return no match
end if
for all $(r, i) \in C$ in order **do**
  in a transaction: re-read i and q_d; if i is still pending and $q_d > 0$, mark i matched, set $q_d \leftarrow q_d - \min(q_d, need_i)$ and stop
end for
create delivery; self-delivery is auto-accepted, otherwise assign the nearest available volunteer (ties: fewest active deliveries)

where $w_{\text{sev}} \in \{40, 30, 20, 10\}$ (critical to low), n_r is the number of people affected, $v_r \leq 4$ counts vulnerable groups (children, elderly, pregnant women, people with disabilities) and h_r is hours waited. Three points need stating. (i) The vulnerable term reaches at most 20, so the cap of 25 in the code never applies. (ii) The score is computed once, when $h_r \approx 0$, so waiting time does not yet affect rank; the rule effectively has *three* active criteria and ranges from 10 to 90. (iii) The people term saturates at 15 people, so a household of 15 and a village of 1,500 score the same. The weights were set by the designers and are not taken from Sphere standards or DMC guidance. Section V-C tests their sensitivity.

C. Matching and Assignment

Algorithm 1 picks the highest-scoring verified request item in the donation's category and uses Haversine distance only to break ties. Each match consumes the smaller of the donation's remainder and the item's need. Volunteers are ranked by distance to pickup and then by active workload. Siblings from one multi-item drop-off are kept with the same volunteer where possible. Each match runs inside a Firestore transaction that re-reads the item and the donation's remainder before writing. If a concurrent match has already taken the item, the next-best candidate is tried, so two matches cannot claim the same item or spend the same balance (Section V-B).

D. Handover Check and Threat Model

When a delivery is marked *delivered*, the server issues a single-use 128-bit token. The token is shown as a QR code

TABLE II
REQUIREMENT EVIDENCE BY FEATURE AREA

Area	FRs	Formal test	Dev. check only	Not tested
A Intake & administration	01–07	6	1	0
B Donation & delivery	08–14	5	1	1 (FR-11)
C Maps & forecasting	15–19	3	2	0
D Emergency oversight	20–24	2	3	0
E Engagement & platforms	25–30	3	3	0
F Inventory & coordination	31–33	2	1	0
Total	33	21	11	1

TABLE III
CONCURRENT-LOAD LATENCY PER RUN (S)

		Run 1		Run 2	
Phase	n	p50	p95	p50	p95
Submit requests	30	5.7	6.1	2.9	3.2
Approve requests	30	9.1	9.5	3.2	3.3
Register donations	30	4.9	5.2	1.6	1.7
Match (race probe)	30	17.9	271.2	10.0	25.5
Volunteer accept	15	3.6	3.8	74.0	74.3
Read burst (cached)	50	0.7	0.7	1.8	3.1

Successful requests only; 100% success in every phase; zero double-matched items; runs six minutes apart.

only to the party holding the goods and is removed from the victim's API responses. The server confirms the delivery only if the caller owns the request and the token matches; otherwise it returns HTTP 403. This prevents (a) one-sided confirmation by a deliverer, (b) confirmation of someone else's delivery and (c) guessing the token. It does *not* prove that goods physically changed hands. A photographed QR code could be relayed remotely, and a colluding victim and volunteer can confirm a delivery that never happened. We therefore treat the scan as evidence of handover (DP2), not proof, and recommend audit sampling. Self-reported severity and vulnerability can also be inflated. Mitigations are administrator verification, duplicate flags on NIC number or address, and an append-only audit log. Verification does not adjust the score, which leaves a gap.

E. Privacy and Other Safeguards

NIC numbers, addresses and exact locations are visible only to their owner and to administrators. Public endpoints return district aggregates (DP3), in line with Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [27]. People without an NIC, such as children or undocumented people, cannot be checked for duplicates by NIC. The SOS module is a supplementary channel. It does not replace official dispatch, carries no guarantee of response and has not been evaluated in a real emergency. The interface does not yet display the official DMC 117 hotline [28] or such a disclaimer; both should be added.

An auxiliary, separately evaluated flood-risk layer (a logistic regression trained on DesInventar and NASA POWER data) is shown on the maps. It never affects allocation.

V. EVALUATION

A. Verification and Traceability

Table II maps the 33 FRs to the strongest evidence available for each. Twenty-one FRs have a passing formal test case, eleven were checked only live during development, and FR-11 (rejection reopening a donation) is implemented but was not separately tested. Evidence is partial for FR-12 (camera round trip not exercised), FR-27 (on-device push not observed) and FR-30 (mobile maps and push not verified on a physical device). The automated suites passed 77/77 unit tests, 48/48 API checks across 14 route files and 45/45 direct Firestore security-rule checks. The rule tests found an over-broad read rule on deliveries, which was corrected and re-verified.

B. Concurrency and Performance

A script created 76 throwaway accounts and fired each phase's requests simultaneously at a single local API instance backed by cloud Firestore. Test requests, donations and volunteers were placed in an offshore area, so that nearest-volunteer assignment always chose a test volunteer; all 15 acceptances in each run went to test accounts. In the race probe, 30 donations competed for 30 water-request items. An earlier version read candidates and wrote without a transaction, a classic lost-update pattern for a partial-quantity balance; no double allocation was observed, but that only showed that the race had not been triggered. Matching therefore now runs in a transaction (Section IV-C), and both runs produced zero double allocations, now guaranteed by design. The cost is contention: retried transactions gave a matching p50 of 10–18 s and a p95 of 25 s in one run and 271 s in the other (Table III), against about 9 s p95 without the transaction. In normal use administrators trigger matches one at a time, so this is a worst case. Latency also varied widely between the two runs, held six minutes apart: run 1's write phases were two to three times slower, and in run 2 every acceptance took about 74 s, a uniform stall that points to throttling of a shared backend resource rather than per-request work; its cause was not identified. With two runs, these figures are indicative only.

The dashboard statistics endpoints scanned whole collections on every call, giving a p95 of about 11 s for 50 concurrent reads. A 30 s cache made this worse (p95 about 25 s) because concurrent cache misses each started their own scan, a cache stampede. Caching the in-flight promise (a single-flight pattern) reduced p95 to about 2.4 s. The remaining 2.4 s is the cost of one full scan on a cold cache. The cache is held in memory, so on serverless hosting each new instance starts cold and instances do not share results. Precomputed aggregate documents or Firestore count queries would remove the scans.

C. Allocation Simulation and Sensitivity

To cover the scenario demonstration (DSRM phase 4), we simulated a 72-hour response with 3,000 requests. Requests were placed by districts' historical flood frequency from DesInventar. Supply was 70% of demand, 60% of donors were in Colombo, Gampaha and Kandy, and matching ran hourly. We ran 30 seeds and compared the deployed rule with first-in-first-out (FIFO), nearest-first and random matching, and

TABLE IV
SIMULATED ALLOCATION (MEAN OF 30 SEEDS)

Policy	Crit. served	Low served	Crit. wait (h, med.)	Mean km	Gini	Worst district
Priority (deployed)	.996	.371	1.3	115	.027	.70
FIFO	.787	.783	28.8	140	.027	.71
Nearest-first	.785	.785	16.5	41	.178	.29
Random	.787	.785	19.7	140	.022	.72
Priority + aging	.994	.379	6.0	123	.027	.70
Priority + 100 km	.982	.493	3.3	63	.127	.43

Overall item fulfilment was 0.78–0.79 for every policy, so supply sets how much is delivered and the policy sets who receives it. "Served" = share of items matched within 72 h. Gini = Gini coefficient of district fulfilment ratios; Worst district = lowest district fulfilment ratio. Across seeds, SD $\leq$ 0.03 for service rates and Gini, and $\leq$ 0.05 for the worst district.

with two variants: aging, which re-scores with the wait term each cycle, and a 100 km distance cap. The simulation models allocation only, not travel or delivery. Table IV reports the results.

Priority-first matching gave critical needs near-complete and fast service. Low-severity needs, however, were mostly left unmet (37% vs. 78%), confirming the risk of starvation. Aging with the current 20-point wait cap did not fix this (38%), because the cap is smaller than the 30-point severity gap between critical and low, and it delayed critical items. Any aging scheme therefore needs a larger or uncapped wait term. Distance broke ties in 72% of matches, and 28% of matches were more than 150 km apart, which is impractical for self-delivery. Nearest-first matching shortened distances but served the worst district only 29% of the time. A 100 km cap halved mean distance but lowered equity across districts (Gini 0.127). This trade-off between feasibility and equity matches [19].

For sensitivity, we varied each weight by $\pm$25–50% on 15,000 synthetic requests. Kendall's τ against the baseline ranking stayed at or above 0.83, and overlap in the top 10% stayed at or above 0.79. In 1,000 joint random perturbations, the median τ was 0.90 (5th percentile 0.80). The score took only 73 distinct values, so ties are pervasive. Replacing the people term with $\min(10 \log_{10}(1 + n_r), 30)$, which saturates near 1,000 people, changed rankings more than any weight change ($\tau = 0.78$, top-10% overlap 0.61). How the people term is scaled is therefore the most consequential open choice and should be checked against expert rankings.

D. Flood-Risk Layer

The model is an L2-regularised logistic regression (batch gradient descent, 3,000 epochs, learning rate 0.3, $\lambda = 0.001$, z-scored features, no class weighting) with five features: monthly and preceding-30-day rainfall, sine and cosine of the month, and the district's historical flood rate. In-sample, it scored AUC-ROC 0.825, with recall of only 12.4% at a 0.5 threshold. The interface does not use that threshold. It shows four risk bands (15%, 30% and 50% cut-offs) as a ranking. On a temporal hold-out (training 1981–2010, testing 2011–2019, district rate computed from training years only), the model reached AUC-ROC 0.761, PR-AUC 0.354, Brier score 0.120 and top-decile precision 0.419 (test base rate 0.146). It outperformed climatology baselines: district$\times$month (AUC 0.687, Brier 0.123), month only (AUC 0.720, Brier 0.121) and a constant (Brier 0.128). An ablation showed that monthly rainfall carries most of the skill (AUC 0.664 without it). The district historical-rate feature hurt generalisation: dropping it raised held-out AUC to 0.787 and lowered Brier to 0.111, so it should be removed. The model is overconfident above a predicted probability of about 0.4 (top bin: predicted 0.95, observed 0.50). This is partly because DesInventar reporting grows sharply after 2002 and almost stops in 2020 (positive rate 0.087 in training, 0.146 in testing). The layer is therefore presented as a ranking aid, not a warning tool.

E. Usability Study

Nineteen participants (4 victim, 5 donor, 5 volunteer and 5 administrator roles), a mixed convenience sample of students, family members, friends and working adults recruited by word of mouth, each completed four scripted think-aloud tasks with seeded accounts in English, so the Sinhala and Tamil interfaces were not tested. The developer facilitated, read each task aloud and performed any back-end steps a scenario required. A task counted as "unaided" when no help was recorded for any of its steps. The SUS score [29] had a mean of 76.7 (SD 11.9, 95% CI 71.0–82.4, range 55–97.5), in the "good" adjective band [30]; internal consistency was high (Cronbach's $\alpha = 0.93$). Role means were similar: victim 73.8, donor 77.0, volunteer 78.0 and administrator 77.5, but with four or five participants per role these differences are not interpretable. Of 76 tasks, 58 (76%) were completed unaided, 14 with help and 4 not at all (completion 94.7%), with 16 errors and a mean time on task of 619 s (SD 109). Help was mostly needed to locate delivery status and dashboard options. Some participants had disaster-response (DMC or NGO) involvement or had been affected by a disaster; to protect anonymity, their number and roles are not reported. Social desirability and familiarity with the developer may have inflated scores.

VI. DISCUSSION

Research question. The artefact *supports* transparency and coordination: allocations can be traced to disclosed inputs (DP1), handovers leave verifiable evidence (DP2), and aggregate status is public (DP3). It does not yet show improved efficiency or equity. Instead, the simulation shows that a disclosed rule makes its trade-offs, such as starvation of low-severity needs and long matching distances, visible so they can be debated and tuned; that is the main practical value of an explainable rule. Compared with Sahana [11], the added value lies in the published rule, the handover check and public aggregates. It does not lie in integration itself.

Engineering lessons. Single-flight caching avoids stampedes but, on serverless hosting, should be backed by precomputed aggregates. Transactions make allocation correct under concurrency, but contention on popular items shows up as tail latency, which a queue per category can absorb.

A. Threats to Validity

Internal: the load tests ran against one local instance, not the serverless deployment, with only two runs; volunteer assignment is isolated from real accounts, but the harness shares the live database and matching selects from all verified requests. The usability sessions were facilitated by the developer. *External:* the simulation's inputs (severity mix, donor concentration, supply ratio) are assumptions. The usability sample is a convenience sample, not representative of affected users or responders, and the rule weights are not validated by experts. *Construct:* "served" measures allocation, not delivered aid. SUS measures perceived usability, not coordination outcomes. Flood labels record reported incidents, not floods that occurred. *Coverage:* the platform needs connectivity and a smartphone. SMS is outbound only; there is no USSD, call-centre or offline QR fallback. Sinhala and Tamil strings are unreviewed machine translations, per-request scoring treats water and clothing as equally urgent, and every request waits for manual administrator verification, which becomes a bottleneck in a surge unless verification is delegated to district officers.

VII. Conclusion

We presented a disaster-aid platform for Sri Lanka built on explicit design principles and evaluated with complementary methods. The disclosed rule served critical needs quickly. Simulation exposed its starvation and distance costs, and testing led to transactional matching and a fix for a cache stampede. Future work comprises: uncapped aging; a distance-feasibility threshold and category-specific urgency; a matching queue to cut transaction contention; validation of the weights through expert ranking with DMC and NGO staff; offline and SMS intake; dropping the district-rate feature and recalibrating the flood model; and a larger, representative usability study before any field pilot.

Data availability: code, test scripts and simulation inputs will be released in a public repository upon acceptance (link withheld for double-blind review).

Use of generative AI: a generative AI tool (Claude, Anthropic) was used to help draft and edit the text, restructure the manuscript, check references, and write the allocation-simulation and model-evaluation scripts. The authors reviewed, verified and take responsibility for all content, data and results.

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